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ORIGINAL RESEARCH

ARDI: a new dataset for automatic advocate recommendation in the Indian Legal System

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Abstract
We introduce a realistic expert recommendation problem called advocate recommendation. To facilitate investigation of the problem, we develop a rich dataset of ∼25k documents called the Automatic Advocate Recommendation Dataset in the Indian Legal System (ARDI), which also contains additional attributes. Extra information about areas is generated through an expert annotation process that we incorporate into our experimentation. Treating the problem as a multi-label classification task and carrying out extensive experimentation with various strategies, including using area-based representations, summarization, ensembling methods and multi-task learning, we find the advocate recommendation task quite challenging. Our results suggest that using longer contexts and combining information from different models are beneficial in reaching a viable solution.

Keywords Advocate prediction · Dataset

1 Introduction

Advocates play a crucial role in the Indian legal system as most parties (petitioner or respondent) are represented by advocates of their choice to argue the case on their behalf Papa and Wilkins (2011) (https://clp.law.harvard.edu/knowledge-hub/magazine/issues/indian-legal-profession/indian-lawyers-and-the-state). Consequently, the selection of appropriate advocates is an important step that can influence the outcome of a case. Therefore, there is a need for a reliable and robust system for recommending advocates automatically.

However, the selection of advocates, in the present Indian legal system, is mostly done by word-of-mouth information and ease-of-access. While the Bar Council of

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India and the respective State Bar Councils maintain an official record of registered advocates, this information is usually difficult to find due to the lack of a uniform data reporting practice. Several states offer services to check the legibility of advocates using their names but not for finding advocates given a situation. Furthermore, Indian laws prohibit advocates from advertising themselves, making it even more difficult to find information about lawyers[1]. The combination of these aspects sets the landscape for looking into what we believe to be a very practical and socially relevant problem: automatic recommendation of advocates. That is, given a factual description of a potential legal dispute, the problem of advocate recommendation is to find competent or well-suited advocates.

A robust system for recommendation of advocates could have several positive impacts on the legal machinery. First and foremost, this would allow clients to make more informed decisions about choosing their representatives. Second, the Indian judiciary system suffers from a severe backlog of cases arising from shortages of judges[2] and an inadequate infrastructure to effectively distribute cases to advocates. An efficient advocate recommendation system would help address part of the backlog and reduce case pendency by streamlining case assignment to well-suited advocates. An efficient case-advocate assignment strategy could similarly be leveraged by private law firms to choose the best set of advocates for a case.

Considering the integral nature of advocates in the Indian legal system, a pressing need for better infrastructure to distribute pending cases, the legally-driven difficulty in finding proper legal representation, and a reasonable expectation for competent legal counsel, advocate recommendation is a highly practical, nuanced and complex problem, a solution to which, would address a key component behind the slow legal machinery in the Indian system. The lack of systematic information about advocates forces methods to recommend advocates to apply roundabout strategies such as extracting advocate names from official court documents, identifying areas of specialization of advocates based on the legal statutes cited in their court documents and applying a mapping of statutes to legal areas obtained through a comprehensive annotation study to facilitate accurate recommendation based on areas of relevance.

A major hurdle in preparing an empirical test-bed for the proposed *advocate recommemdation* task is the absence of a suitable dataset which has the relevant advocates annotated for a set of facts. Therefore, we formulate the practical problem of advocate recommendation, as an expert recommendation task that contains several data constraints (Section 3). To evaluate the difficulty of the task, we develop a rich dataset of ∼25K documents for the task of advocate recommendation called the **Automatic Advocate Recommendation Dataset in the Indian Legal System (ARDI)** (Section 4), *which we will release upon the acceptance of the paper*. The problem of advocate recommendation is treated as a multi-label classification task where the input is the *facts* of a court case and the output is the *advocates* corresponding to that case. We carry out extensive experimentation based on different strategies that we discuss in Section 5.

[1] Rule 36 at http://www.barcouncilofindia.org/wp-content/uploads/2010/05/BCIRulesPartVonwards.pdf

[2] "The clogged state of the Indian judiciary"; https://www.thehindu.com/news/national/indian-judiciary-pendency-data-courts-statistics-explain-judges-ramana-chief-justiceundertrials/article65378182.ece

To the best of our knowledge, the task of advocate recommendation in the Indian legal setup has not been attempted before. Therefore, our work makes the following contributions:

- We propose the task of advocate recommendation, a socially relevant and practical task for legal AI in the Indian scenario.
- We provide a new dataset called the ARDI dataset for advocate recommendation (Section 4). The ARDI Dataset contains information for 306 advocates and per-document statute information spanning 5555 Sections, 653 Chapters and 351 Acts. In addition, it also contains expert-annotated area information across 36 Areas that add an additional layer of categorization for the statute information present. The dataset also includes per-sentence rhetorical role information (Deroy et al. 2021) with 16595 factual sentences (Bhattacharya et al. 2019) in the working corpus. Lastly, the ARDI dataset provides case verdict information for all cases.
- We highlight the challenging nature of the advocate recommendation task through extensive testing with SOTA models and other model architectures (Section 5). Our work establishes a broad set of strong baselines for the task (Section 6.3). We also provide an analysis of the present challenges of the task and possible future improvements (Section 7).

2 Problem

The requirement for legal counsel begins when a potential client believes a situation they experienced requires legal intervention or, a court of law has to assign legal representatives to one or both parties involved in a legal dispute. Thus, **the problem of advocate recommendation is to find competent or well-suited advocates when presented with a factual description of a potential legal dispute**.

We formulate the problem of advocate recommendation as a multi-label classification task, considering that certain cases require several advocates representing each side. We begin with presenting a simple formulation of the advocate recommendation task below followed by an additional formulation that clarifies how relevant facets of the ARDI dataset influence the evaluation objective.

Let F be a textual description of a potential legal dispute that we call a *Fact* (Bhattacharya et al. 2019).

Let $\mathcal{A} = \{A_1, A_2, \ldots, A_m\}$ be a set of candidate advocates to recommend from. The general task of advocate recommendation is defined as:

$$\text{Recommendation}(\text{F}, \mathcal{A}) = \{A_i | A_i \in \mathcal{A};\ A_i \text{ well-suited to F}\} \tag{1}$$

Where the criterion 'well-suited to' is an abstract criteria that measures how well-suited an advocate is for a particular Fact. On the ARDI dataset, the availability of the actual counsel advocates replaces this criteria with the ground truth.

The ARDI dataset has been curated with gold standard advocate targets for cases to assess the ability of advocate recommendation techniques as outlined in (1). The

following formulates the task of advocate recommendation and its evaluation on the ARDI dataset.

Let $\mathcal{C} = \{C_1, C_2, \ldots, C_n\}$ be cases of a court of law (from the ARDI Dataset).

Let $\mathcal{F} = \{F_1, F_2, ..., F_n\}$ be the corresponding set of *Facts* (Bhattacharya et al. 2019) extracted from each case such that F_i is the Fact description of case $C_i \in \mathcal{C}$.

Published case texts contain additional information (rhetorical roles (Bhattacharya et al. 2019)) such as verdicts and reasoning by both parties (Deroy et al. 2021) that would not be available in a real scenario. As the formulation of advocate recommendation represents a realistic scenario where only Facts would be present, our formulation focuses only on utilizing Facts for advocate recommendation.

Let $\mathcal{A} = \{A_1, A_2, \ldots, A_m\}$ be the extracted set of advocates appearing as counsel for the cases in $\mathcal{C}$.

Let $A_{C_i} \subset \mathcal{A}$ be the advocates appearing as counsel in case C_i. The objective of advocate recommendation on the ARDI dataset is then defined similar to (1) as:

$$\text{Recommendation}_{\text{ARDI}}(F_i, \mathcal{A}) = \{A_k | A_k \in A_{C_i}\} \tag{2}$$

3 Related works

Expert recommendation To the best of our knowledge, Askari et al. (2022) is the only similar work that addresses legal expert recommendation. Treated as an information retrieval task, a question-answering dataset has been created from an online legal forum and is used to generate a ranked list of advocates for new queries. While the legal expert identification problem has been attempted here, we find it to be sufficiently different from our setup. In particular, the work treats the problem as an information retrieval problem and uses a question-answering dataset, and leverages extra information about the experts. This differs from our multi-label classification approach where the choice of a fact-based dataset, with no additional information about experts, makes it difficult to compare.

Doctor recommendation in clinical trials using a dynamic databank of patient representations and additional doctor information is carried out in Biswal et al. (2020). We experiment with a similar setup 5.5) and compare its performance with other strategies (Section 6.3). As opposed to doctor recommendation as addressed in their work, advocate recommendation is a more constrained recommendation problem as we do not possess extra information about our experts. In addition, we work with extracted sentences as opposed to well-crafted descriptions in Biswal et al. (2020).

Legal statute prediction As we conduct experiments with area prediction, we make a note of a few works from the closely related area of Legal Statute Prediction. Paul et al. (2020) and Paul et al. (2022a) work on charge prediction at the level of chapters on Indian Supreme court data. In their work, Luo et al. (2017) try to find out whether charge prediction models are able to understand the legal theories behind the charge prediction task, foraying into understanding the reasoning capabilities of such mod-

els. Paul et al. (2024) presents a detailed study of the legal statute identification task on standard datasets (Indian and European and methods (mostly transformer-based).

Legal datasets and other legal tasks We are not aware of any dataset that addresses the problem of advocate recommendation as we have. However, due to its closeness to the tasks of statute and charge prediction, we discuss other datasets dealing with Indian legal AI tasks. A small dataset of 50 legal documents for rhetorical role identification has been provided in Bhattacharya et al. (2019). Bhattacharya et al. (2021) and Nallapati et al. (2017) provide gold-standard summaries of legal documents that can be used for training legal summarization strategies. Paul et al. (2020), Paul et al. (2021) provide a large dataset for the task of statute/charge prediction in the Indian context. In judgment prediction, Malik et al. (2021) provides a dataset of 34816 documents, namely the Indian Legal Document Corpus annotated with judgment labels for every legal document. Nigam et al. (2024) presented the largest expert-annotated dataset for legal judgment prediction and explanation in the Indian context, with benchmarking results with state-of-the-art models.

Legal Recommendation Since the present work is on recommendation in the legal domain, we skim through some other works on recommendation in the same domain. Wang et al. (2024) leveraged a Bayesian network in legal recommendation producing notable performance esp. for sparse data. Yang et al. (2021) proposed a graph neural network-based recommendation system in "Legal-Rec" scenario to help users fetch legal documents. Huang et al. (2021) propose a deep learning based citation recommendation system using a citation-list based method and context-based methods.

4 Dataset

To evaluate the task of advocate recommendation, we develop a dataset using cases from the Delhi High Court which consistently records the names of advocates appearing for both parties in the case texts. We gathered 25640 Delhi High Court case documents from **www.indiankanoon.org** ranging in the period of June 2008 to March 2022. We name our dataset the **Automatic Advocate Recommendation Dataset in the Indian Legal System (ARDI)**.

In addition to the petitioner and respondent counsel information, the ARDI dataset also provides information about the rhetorical roles of sentences in the case texts, case verdicts and extracted statutes at all levels of granularity within the Indian legal system, including an additional area-level granularity for statutes that we developed through an annotation study. Section 4.1 outlines the method by which the 25640 raw HTML documents were processed to obtain the final usable dataset that is utilized for all our experimentation in Section 5. In Section 4.2, we outline the area annotation study that was undertaken to further enrich the statute information of the dataset and that comprises the primary motivation of our method outlined in Section 5.3.2. Section 4.3 provides a comprehensive overview of the dataset emphasizing its usability for advocate recommendation. It provides an overview of the design considerations

made for the ARDI dataset when performing experiments along with advocate concurrency information in cases to highlight the complexity of the problem.

4.1 Dataset curation

We outline the process by which the 25640 raw HTML court documents is converted into a rich dataset for advocate recommendation and other legal AI tasks. Figure 1 provides an overview of the structure of the original HTML documents of the cases and the attributes we extract. Before extracting these attributes, we undertake an initial pre-processing of the raw HTML documents. This involves removing non-UTF-8 characters, repeated punctuation marks, and substitution of legal-specific abbreviations with their full forms e.g. 'ss' with 'sections' that aid us later in extracting statutes.

4.1.1 Advocate name extraction

The names of the advocates are mentioned in the preamble portion of each document which spans either the <pre> tag or the first four paragraphs (<p> tag). Figure 1 provides an example of the information contained in the <pre> tag. Keywords like

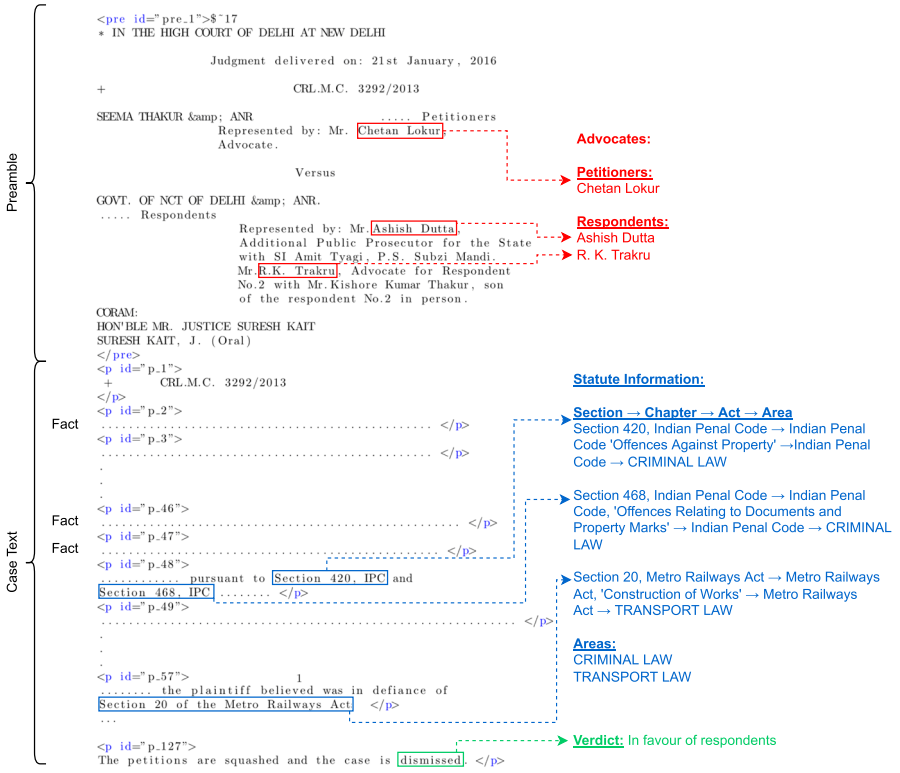


Fig. 1 Example of HTML document of a case and relevant information present in the document

'Petitioners', 'Plaintiffs' and 'Respondents' preface the names of advocates in the documents. Using regular expressions that leverage these keyword-based patterns (exact expressions in Table 1), we extract the names of all advocates featured in the document and the names of petitioning advocates due to more consistent reporting styles. The names of the respondent advocates are obtained as the set difference between the two groups of advocate names extracted. Through this process, we extracted 21531 advocates from the corpus. However, a large number of these advocates are associated with only a single case. As a result, for this work, we choose to work with only those advocates having at least 30 cases. Figure 2 provides a histogram showing advocate distribution over different numbers of cases.

4.1.2 Court verdict extraction

Terms like 'upheld' and 'discarded' imply rulings in favour of the petitioners and respondents respectively. Using such indicative patterns, we identify the winning sides of each case. While we do not make use of the verdicts of cases in our present work, we include it as part of our dataset and look to incorporate it in subsequent works.

Table 1 Regexes used to extract names of all advocates and of Petitioners (implemented in a case-insensitive manner) (broken into several lines to fit page-width)

Advocate Type	Regular Expression
All	`(?:D|M)(?:r|s|rs)_*[A-Za-z.]++[A-Za-z.]+(?:*[A-Za-z]+)?,?`
Petitioners	`(?<=(Petitioner|Apellant|Appellant|Plaintiff)s?)(?P<pet>.*?)versus.*?(?=Respondents?)"(?P<res>.*?)`
Petitioners (alternate)	`(?<=For the (Petitioner|Apellant|Appellant|Plaintiff)s?)(?P<pet>.*?)versus.*?(?= For the Respondents?)"(?P<res>.*?)`

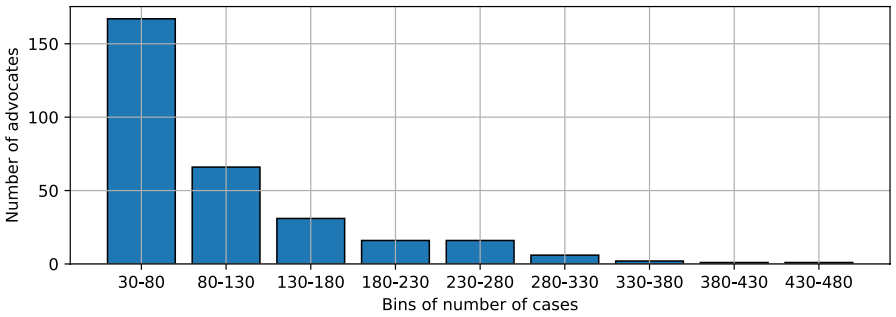


Fig. 2 Advocate distribution over different number of cases. x-axis represents the number of cases in which an advocate appears. y-axis represents the number of advocates appearing in this number of cases. E.g. the bin 30–80 (x-axis) and 167 advocates (y-axis) show that each of these 167 advocates contains 30–80 cases

4.1.3 Section extraction

The laws of India are laid down in Acts. These legal documents have a three-level hierarchy going from fine-grained to coarse-grained in their scope as follows: **Sections→Chapters→Acts** (Paul et al. 2020). To get all relevant information in this hierarchy, we use the section extractor in Podder and Bhattacharya (2020) to obtain the sections from each document. With the section information for each case, we map the cases to their relevant Chapters (if they exist) and their Acts. This multi-granularity statute information about the cases makes it a valuable dataset for other legal tasks like statute prediction.

4.1.4 Sentence segmentation, rhetorical role classification and masking

As outlined in Section 2, we utilize only the Facts of each case text for advocate recommendation. In a case document, facts are the mentioned events that lead to the filing of a case. To obtain the facts from each document, we utilize the rhetorical role sentence classifier from Bhattacharya et al. (2019) to automatically label each sentence of a case document according to one of seven rhetorical roles. We devise the Fact of each document by sequentially combining all sentences labelled as *Facts* by the classifier.

Once the facts are obtained from the documents, we **mask** the names of all named-entities including the names of advocates and sections appearing in each sentence to ensure no information provides any model an unfair advantage (we use the named-entity tagger from the spacy toolkit).

4.1.5 Advocate statute information

Using the extracted Section information of the cases (Section 4.1.3), we collate Section, Chapter and Act statistics of each advocate from the cases they appear in. This information forms the backbone of the area-contextualized experimentation we discuss in Section 5.3.2 and for the area prediction portion of the task in Section 5.4.

4.2 Area annotation

Our discussions with legal experts revealed that the areas of expertise of advocates are more coarse-grained than the Act level of the hierarchy we mentioned in Section 4.1.3. An example would be the Indian Penal Code of India and the Code of Criminal Procedure belonging to the area of 'CRIMINAL LAW'. Finding such areas for the cases could help distinguish cases along their areas and might be leveraged to help models improve their recommendations. Therefore, we embarked on the task of obtaining coarse-grained area information about Acts. Two legal experts (an LLB. student and a Law faculty member, both Indian nationals) worked in sequence to provide annotations for the Acts. The annotation process was carried out in three stages.

1. The faculty member **created an exhaustive set of legal areas**. This yielded a total of 40 areas.

2. The LLB. student was provided with a list of all central Acts of India along with the list of areas. The annotator performed the task of **annotating each Act with one or more areas**. Aggregating the information highlighted 36 of the 40 areas as relevant to the provided list of Acts.
3. The faculty member corroborated the annotations made by the student and the two engaged in discussions to **jointly resolve any discrepancies**.

We go into further detail about the area-based imbalances in the dataset as well as other attributes in Section 4.3.

4.3 Dataset statistics

Of the total 21531 advocates that were extracted from 25641 legal documents, a large number (10685 advocates) appear only in a single case. As this is insufficient data to train various models, we select only those advocates having at least 30 cases in the dataset. This yields 306 advocates. We consider only the cases in which these advocates appear as part of our dataset to carry out experimentation. Further, we only retain those cases in which at least five sentences were labeled as Facts to provide sufficient information for models to train with. This leaves us with a working corpus of 16595 case facts (Tables 2, 3, and 4). We split the set of documents in a 70:10:20 split for the training, validation and test splits respectively, and then balance the splits to ensure the representation of all advocates in each split. Table 5 provides the numerical statistics of the working corpus. While the average number of advocates in a case is 1.89, we find that the maximum number is as high as 31 per case. Similarly, while the average number of areas in a case is much lower, there is a very large range with a maximum of 20. These statistics highlight the challenging nature of the multi-label classification tasks that we perform as part of our problem. Tables 3 and 4 provide an exhaustive overview of the frequencies of Areas and Acts in our dataset. Figure 3 provides a schematic overview of the constituents of the ARDI dataset.

Of the 306 advocates, the largest number of cases belonging to a single advocate is 454. The advocate with the least number of cases has 34 cases in total. Based on these disparities, we look at the imbalance ratios (Charte et al. 2013) and concurrence (Charte et al. 2014) for the advocates and for the areas. Table 6 shows the imbalance ratios and the concurrence levels for advocates and areas. Introduced in Charte et al. (2013), *Imbalance Ratio per label*, *Maximum Imbalance Ratio (MaxIR)* and *Mean Imbalance Ratio (MeanIR)* aim to capture the imbalance between labels in the dataset. While the mean imbalance in terms of advocates is not very high, the imbalance is very pronounced in terms of areas. Concurrence of labels can have a significant impact on the performance of a multi-label classification model. *SCUMBLE* (Charte et al. 2014) highlights the average concurrence between imbalanced labels in a dataset. Producing a score between 0 and 1, a higher *SCUMBLE* score represents greater concurrence and therefore, greater difficulty. SCUMBLE scores show significant levels of concurrence for advocates as well as areas. We do not take special measures to mitigate the effects of information imbalance and concurrence for advocates and areas in the present study. The effect of data augmentation for low-resource advocates and areas is an interesting area to explore and we wish to pursue this in future works.

Table 2 Number of cases for the top 40 Acts in the dataset (in descending order of the number of cases)

Act	No. of cases
Indian Penal Code, 1860	4546
Code of Criminal Procedure, 1973	4277
Constitution	3272
Code of Civil Procedure, 1882	1574
Arbitration and Conciliation Act, 1996	910
Indian Evidence Act, 1872	693
Arms Act, 1959	472
Companies Act, 2013	455
Land Acquisition Act, 1894	454
Arbitration Act, 1940	327
Specific Relief Act, 1877	311
Limitation Act, 1963	306
Negotiable Instruments Act, 1881	299
Transfer of Property Act, 1882	293
Prevention of Corruption Act, 1988	262
Indian Contract Act, 1872	256
Delhi Rent Control Act, 1958	169
Customs Act, 1962	158
Industrial Disputes Act, 1947	147
Trade Marks Act, 1999	145
Copyright Act, 1957	119
Delhi Municipal Corporation Act, 1957	107
General Clauses Act, 1897	107
Hindu Marriage Act, 1955	105
Motor Vehicles Act, 1939	87
Indian Succession Act, 1925	85
Contempt of Courts Act, 1971	84
Delhi Development Act, 1957	77
Patents Act, 1970	70
Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013	68
Registration Act, 1908	66
Court Fees Act, 1870	66
Administrative Tribunals Act, 1985	65
Narcotic Drugs and Psychotropic Substances Act, 1985	64
Dowry Prohibition Act, 1961	62
Delhi Land Reforms Act, 1954	62
Delhi School Education Act, 1973	52
Hindu Succession Act, 1956	52
Delhi High Court Act, 1966	51

At present, the treatment of the advocate recommendation task as a multi-label classification problem enables us to be able to predict several advocates for each case. As such, the presence of multiple advocates in each case does not pose a problem by virtue of the problem formulation.

Table 3 Number of cases for each area in the dataset (in descending order of the number of cases)

Area	No. of cases
CRIMINAL LAW	5753
CONSTITUTIONAL LAW	3272
CIVIL LAW	1825
JUDICIARY AND COURTS LAW	1570
PROPERTY LAW	745
NATIONAL SECURITY LAW	609
INTELLECTUAL PROPERTY LAW	592
CONTRACT LAW	574
COMPANY OR BUSINESS OR CORPORATE LAW	496
TRANSPORTATION LAW	454
BANKING AND FINANCE LAW	445
TAX LAW	283
FAMILY LAW	261
GOVERNMENT LAW	223
HUMAN RIGHTS LAW	216
ELECTION LAW	207
ADMINISTRATIVE LAW	113
INTERPRETATION OF STATUTES	108
EDUCATION LAW	106
INTERNATIONAL LAW	94
MOTOR VEHICLES LAW	88
MEDICAL AND HEALTHCARE LAW	75
TRADE LAW	65
IMMIGRATION LAW	63
ENERGY LAW	49
TELECOMMUNICATION LAW	40
RTI LAW	35
CYBER LAW	34
INSURANCE LAW	32
COMPETITION LAW	32
ENVIRONMENT LAW	29
MEDIA LAW	20
CONSUMER LAW	19
ESTATE PLANNING (WILLS AND TRUSTS) LAW	18
AGRICULTURE LAW	16
AVIATION LAW	16
MINING, PETROLEUM AND NATURAL RESOURCES LAW	10

Table 7 highlights that only around 30% of the total number of sentences in the 16595 document corpus are actually Facts. Looking at the number of Fact sentences, we find that the average number of sentences in the Facts of a case is 83. These Facts can be quite long (up to 1499 words). We find the average number of words to be approximately 1200. This word length therefore greatly exceeds the token limit of several neural classification models and we conduct extra experimentation to investigate the need for using greater context in Section 5.1.

Table 4 Number of cases for the top 40 Acts in the dataset (in descending order of the number of cases)

Act	No. of cases
Indian Penal Code, 1860	4546
Code of Criminal Procedure, 1973	4277
Constitution	3272
Code of Civil Procedure, 1882	1574
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Land Acquisition Act, 1894	454
Arbitration Act, 1940	327
Specific Relief Act, 1877	311
Limitation Act, 1963	306
Negotiable Instruments Act, 1881	299
Transfer of Property Act, 1882	293
Prevention of Corruption Act, 1988	262
Indian Contract Act, 1872	256
Delhi Rent Control Act, 1958	169
Customs Act, 1962	158
Industrial Disputes Act, 1947	147
Trade Marks Act, 1999	145
Copyright Act, 1957	119
Delhi Municipal Corporation Act, 1957	107
General Clauses Act, 1897	107
Hindu Marriage Act, 1955	105
Motor Vehicles Act, 1939	87
Indian Succession Act, 1925	85
Contempt of Courts Act, 1971	84
Delhi Development Act, 1957	77
Patents Act, 1970	70
Right to Fair Compensation and Transparency in Land Acquisition,	
Rehabilitation and Resettlement Act, 2013	68
Registration Act, 1908	66
Court Fees Act, 1870	66
Administrative Tribunals Act, 1985	65
Narcotic Drugs and Psychotropic Substances Act, 1985	64
Dowry Prohibition Act, 1961	62
Delhi Land Reforms Act, 1954	62
Delhi School Education Act, 1973	52
Hindu Succession Act, 1956	52
Delhi High Court Act, 1966	51

Table 5 Dataset statistics for the working corpus

Attribute	Total no.	Avg. no./document	Max. count/document
Documents	16595	N/A	N/A
Advocates	306	1.89	31
Areas	36	1.59	12
Acts	351	1.98	20
Chapters	653	3.54	40
Sections	5555	5.27	608

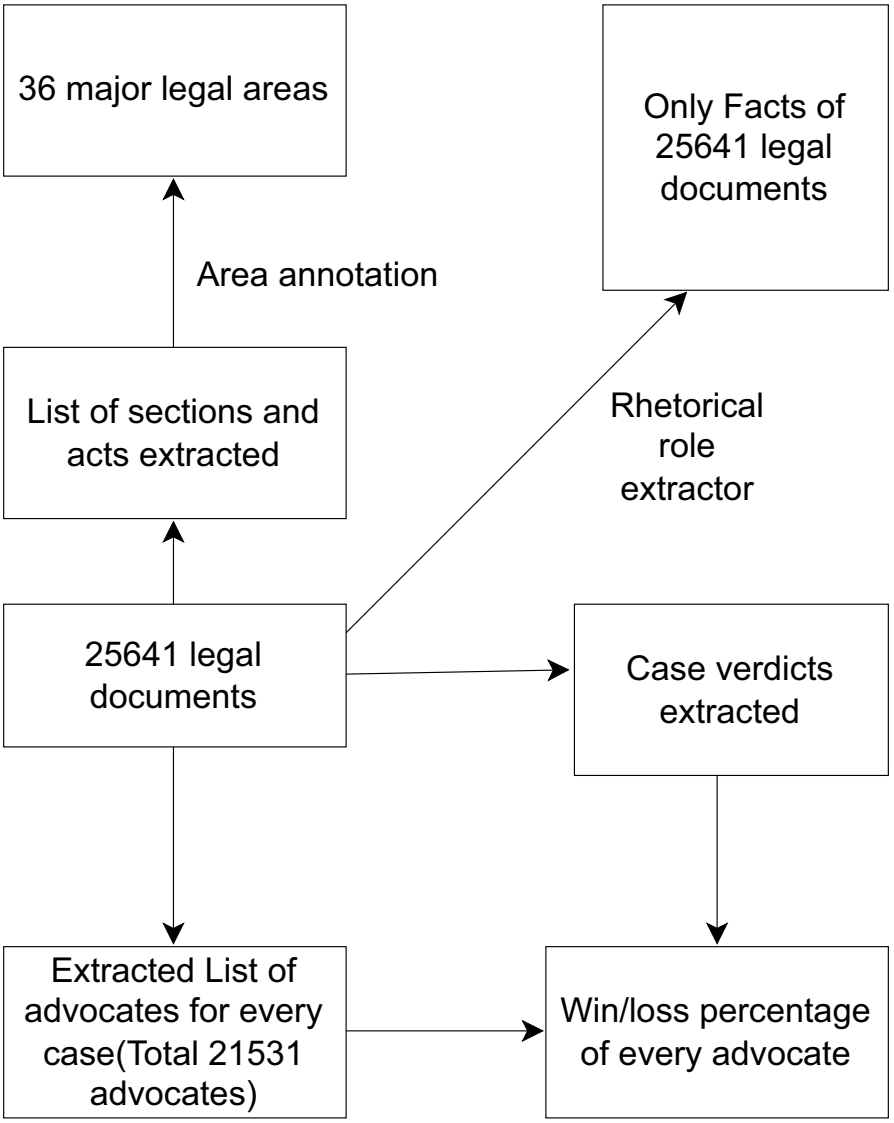


Fig. 3 Constituents of ARDI dataset

Table 6 Imbalance ratios and concurrence levels for advocates and areas

Label Type	*MeanIR*	*MaxIR*	*SCUMBLE*
Advocates	6.09	13.32	0.6793
Areas	99.91	383.34	0.4988

MeanIR is the mean imbalance ratio per label. *MaxIR* is the maximum imbalance ratio. *SCUMBLE* is the concurrence level

Table 7 Fraction of Facts, average number of words and sentences per document for training, testing and validation splits of the dataset

Attribute	Train	Test	Validation
Fraction of Facts	0.3384	0.3518	0.3366
Avg. no. of words/document	1166.06	1233.31	1129.85
Avg. no. of sentences/document	83.26	87.43	80.12

5 Methods

We carry out extensive experimentation on the task of advocate recommendation using different strategies. Each strategy is selected based on investigating the effects of a particular phenomenon on the main task. Our experiments can be categorized as:

- Direct Prediction (Section 5.1),
- Summary-based Direct Prediction (Section 5.2),
- Representation-based Prediction (Section 5.3 along with Sections 5.3.1 and 5.3.2) and,
- Advocate-Area Joint Prediction (Section 5.4).

5.1 Direct prediction

In order to investigate the ability of models to predict the relevant advocates, we first devise experiments to look at the performance of direct prediction by fine-tuning pre-trained Transformer-based models (Vaswani et al. 2017). The various models we experiment with can be considered as *Embedding Generation Models* that encode the fact texts and generate embedding representations.

We consider **BERT** (Devlin et al. 2019), **LEGAL-BERT** (Chalkidis et al. 2020) (a BERT variant pre-trained on European Legal corpus) and **InLegalBERT** (Paul et al. 2022b) (a BERT variant pre-trained on Indian Legal corpus), **RoBERTa** (Liu et al. 2019) and **XLNet** (Yang et al. 2019) for the task of direct advocate prediction.

The length of the fact portions often exceeds the 512 token limitations of the BERT-based models. Therefore, we investigate the effect of greater context length by using a **Longformer** that allows us to work with 4096 tokens and accounts for 98% of the textual data in our dataset.

Lastly, we also considered a hierarchical sequence-based model: **Hierarchical Attention Networks (HAN)** (Yang et al. 2016) which can accommodate all sequence lengths.

5.2 Summary-based direct prediction

Due to the 512 token limitation of several BERT-based models, we perform experiments to investigate the effects of using targeted extractive legal summaries of the facts for the direct prediction task. Such summaries might capture important facets of the Facts that could lead to improved recommendation (Lima and Santos 2022). We work with 4 legal domain-specific extractive summarization methods that we use to generate summaries from the Facts: **CaseSummarizer** (Polsley et al. 2016), **LetSum** (Farzindar and Lapalme 2004), **KMM** (Saravanan et al. 2006) and **SummaRuNNer** (Nallapati et al. 2017) for generating summaries of the facts of a legal case which is then fed into the models as input.

5.3 Representation-based prediction

Using contextualized representations of texts can prove to be an effective way of achieving better performance (Zhang et al. 2020). Askari et al. (2022) and Biswal et al. (2020) both highlight the benefits of using contextualized representations in the expert recommendation. Good representations that distill and encapsulate vital information can eliminate noise present in texts and allow models to focus on important information resulting in better recommendations. Based on this understanding, we perform experiments using two different types of contextualization that we feel might lead to an improvement in performance. We explore the effects of using representations in two ways: **Advocate-contextualised Representation-based** and **Area-contextualised Representation-based** prediction.

5.3.1 Advocate-contextualised representation-based prediction

As our primary objective is to predict advocates, using representations that focus primarily on advocate information of the cases would certainly help improve the quality of predictions. The outputs from the *Embedding Generation Model* (mentioned in Section 5.1) of the various trained Direct Prediction (Section 5.1) and Summary-based Direct Prediction (Section 5.2) models can be viewed as advocate-contextualised embeddings of the Fact inputs. When trained on the task of advocate prediction, the *Embedding Generation Model* focuses on generating representations of the text that maximize the probability of the correct prediction. As a result, these embeddings across all the models carry valuable information about the relevant advocates. When ensembled together, this might lead to an improvement in performance over the individual methods. To explore the effects of combining the representations of these models, we concatenate their representations into a single representation which is then used for predictions. We call this model implementation **Concatenation**.

5.3.2 Area-contextualised representation-based prediction

The second form of contextualized information that we work with is areas of cases. As mentioned in Section 4.2, our discussions with legal experts highlighted that advocates have areas of specialization. Therefore, an advocate is more likely to take

up cases in their area of expertise. This motivates leveraging the area knowledge of cases as a means of getting more distilled information. The area information of cases belonging to different advocates can help discern cases based on their differing areas. As an example, a case that comes under the area of 'TRANSPORTATION LAW' may be better suited to an advocate with expertise in that area. Training a model to identify these patterns might make it easier to make correct predictions about the advocates. Models with the ability to distinguish cases of different areas can then focus on learning the differences between advocates within the same area. Thus, we develop an experiment where we first generate area-contextualized representations of facts and then utilize them to predict advocates.

Figure 4 highlights the overall schematic for the Area-contextualised Representation-based Prediction setup. First, area prediction is carried out from which area-contextualized representations are obtained that are used in the subsequent task of advocate prediction. We detail the three-step process used to carry out area representation-based prediction:

A model is first trained to perform **multi-label area prediction**. For this, we use the same architecture as for the Direct Prediction models outlined in Section 5.1, instead training it using areas as targets. Our choice for the *Embedding Generation Model* is Longformer. This choice is based on other works in the closely related field of legal statute prediction noticing the importance of using greater context (Paul et al. 2020, 2021). We also experiment with generating embeddings using models specifically designed for that purpose. **Sentence-bert (SBERT)** (Reimers and Gurevych 2019) is an embedding generation model that is trained using similarity learning and has been observed to generate better embeddings than BERT. We train an SBERT model on the similarity learning task of areas. More specifically, the model is trained to predict the Jaccard similarity between the areas of two cases, given their facts as input. The trained area prediction/similarity model is then used to **generate area-contextualised representations** of the facts. With these embeddings, we finally **train a simple linear model** to perform the primary task of multi-label advocate prediction.

5.4 Advocate-area joint prediction

The strong connection between the areas of a case and the areas of advocates suggests a possible joint task-based method of predicting advocates. While in Section 5.3.2 we utilized area-based information leveraging area-contextualized representations, the representations were static in nature. Allowing models to jointly learn the areas of a case while predicting advocates would allow the model to dynamically adapt the representations leading to an improvement in performance over using static area-based information. To investigate the effects of using a multi-task setup, we construct a model that jointly predicts the areas of a case as well as recommends its advocates. We define the loss functions for the main task of advocate prediction and the auxiliary task of area prediction using (3), (4), and (5). Let L_{adv} ((3)) and L_{area} ((4)) represent the loss function expressions for the primary and auxiliary tasks respectively.

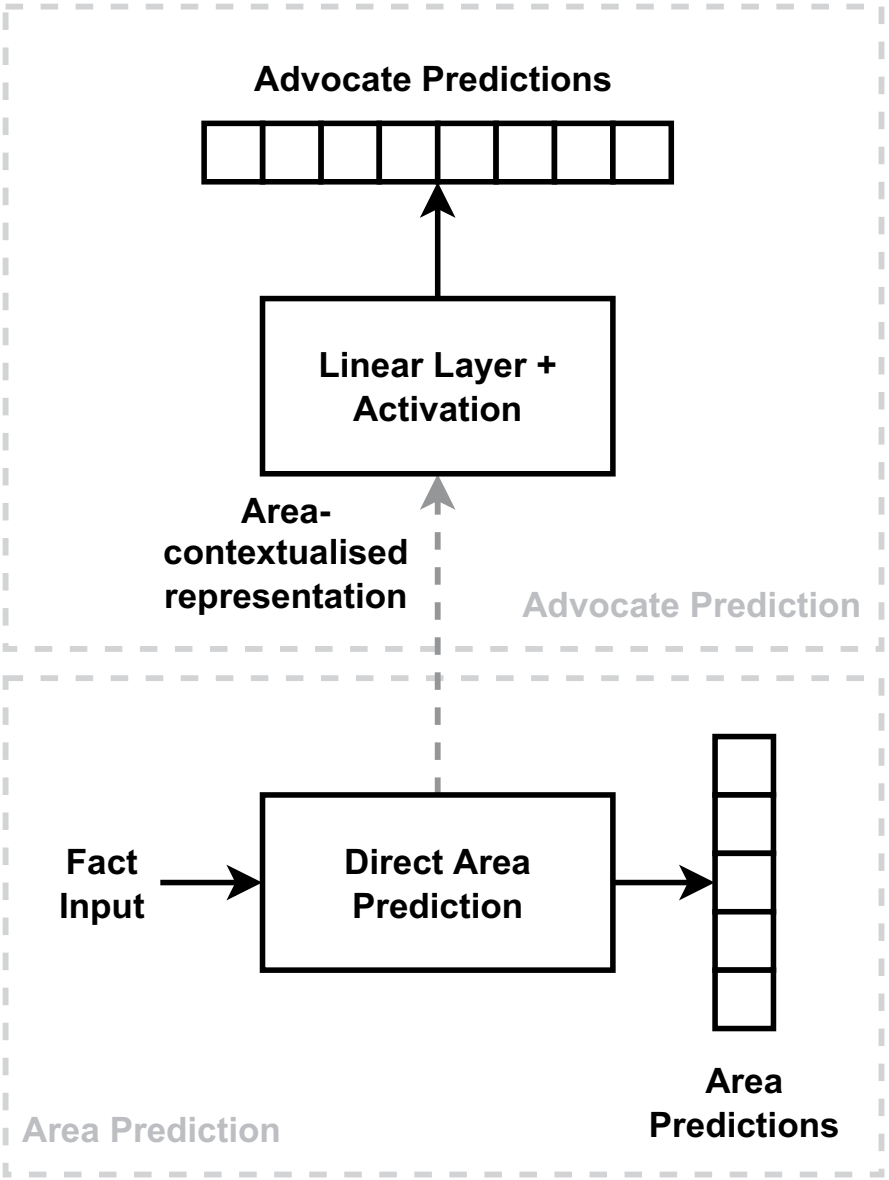


Fig. 4 Area-contextualised representation-based Advocate Prediction

$$L_{adv} = -\frac{1}{N_1}\sum_{i=1}^{N_1}\Big[\, y_i log(P(y_i)) \;+ (1-y_i) log(1-P(y_i))\Big] \tag{3}$$

$$L_{area} = -\frac{1}{N_2}\sum_{i=1}^{N_2}\Big[\, z_i log(P(z_i)) \;+ (1-z_i) log(1-P(z_i))\Big] \tag{4}$$

$$L_{total} = L_{adv} + L_{area} \tag{5}$$

As both tasks have a multi-label classification objective, they use a Binary Cross Entropy (BCE) loss for each label with the total loss (for each task) being the sum of the BCE losses over the labels. y_i and z_i represent the ith advocate and area respectively. N_1 and N_2 represent the number of advocates and areas respectively. The total loss L_{total} is then the sum of L_{adv} and L_{area} as given in (5).

Our hope is that better information separation provided by the auxiliary area prediction task would help the model make better predictions for the primary task. The dynamic nature as opposed to the static area information provided by the embeddings in Section 5.3.2 would provide an additional advantage to learn how to incorporate the area information better for the primary task. We provide a schematic of the multi-task learning model in Fig. 5 . As before, the *Embedding Generation Model* that we employ is Longformer.

5.5 Databank-based prediction

We look to implement the dynamic databank-based prediction setup outlined in Biswal et al. (2020) to evaluate how other expert recommendation strategies perform on our task. The **doctor2vec** model makes use of a dynamic databank of patients and an attention-contextualised representation of the doctors to carry out prediction. For our implementation, we are required to make certain changes due to the lack of certain additional information. The original model uses additional doctor information which we do not possess for the advocates and therefore do not incorporate. Also, we work with static representations as opposed to dynamic representations due to computational constraints. For the representations used (for the static databank and for cases to recommend advocates for), we utilize the SBERT area-based representations that we generated in Section 5.3.2. We randomly choose 40% of the cases from our original training set to be part of the databank with the rest forming the training set.

6 Experimental details

6.1 Performance metrics

As we perform multi-label classification for advocate as well as area prediction, we evaluate the models' performance using the **macro Precision, Recall, and F1 scores**.

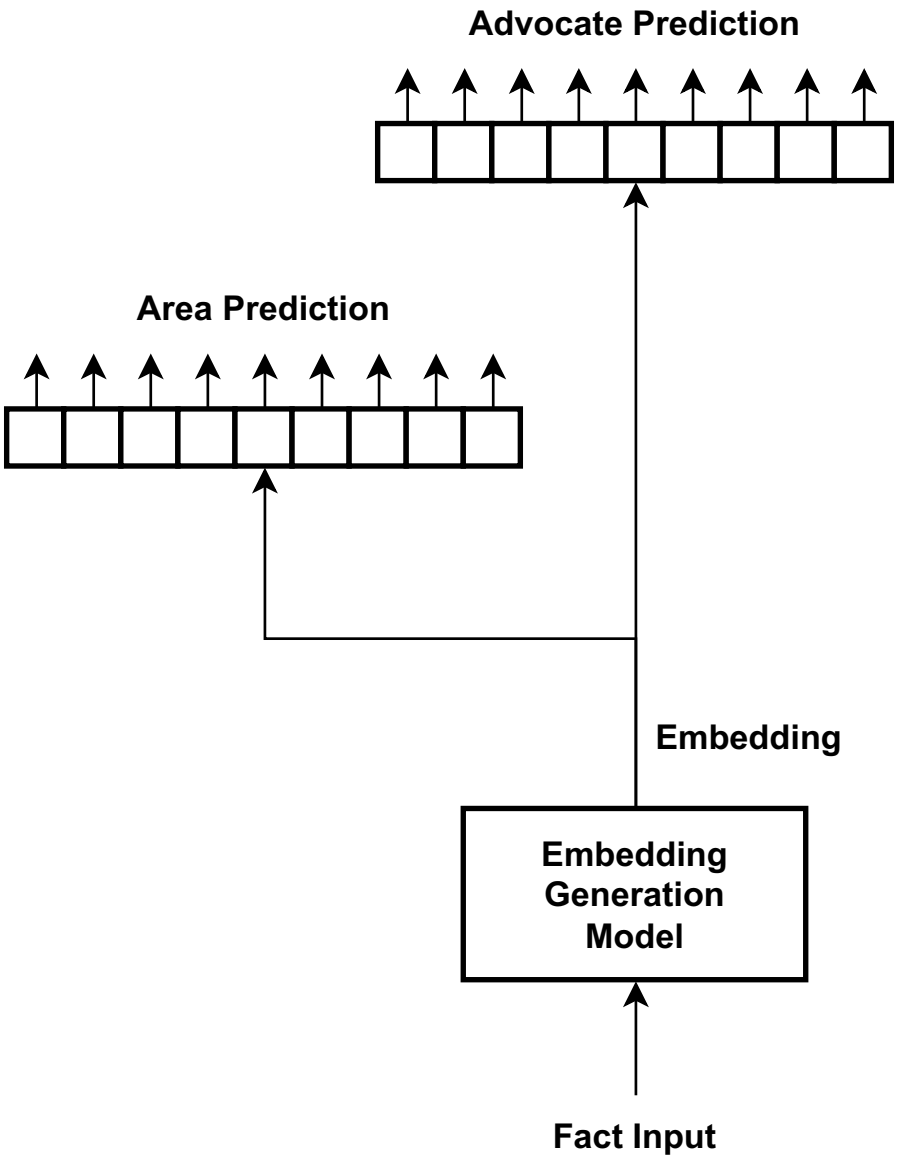


Fig. 5 Advocate-Area Joint Prediction

6.2 Implementation details

All models are implemented in Python using **PyTorch** (Paszke et al. 2019). For the pre-trained language models we use published versions on HuggingFace using the **transformers** module (Wolf et al. 2020). We fine-tune/train all models using the standard recommended set of hyper-parameters and train for 100 epochs. Of the various ways of combining the two losses in the joint prediction setup, we found a simple

addition of the two quantities to yield the best performance. We generate summaries of length closest to the 512 token limitations.

We used one NVIDIA RTX A5000 GPU for running our models. For more details on the model parameters, please refer to Table 8.

6.3 Results

Table 10 shows the results of the best performing models for each category (Table 9 provides results of all models). All models find the task of predicting advocates to be quite challenging.

We observe that ensembling the advocate-contextualized information across different models performs the best. This suggests that the various aspects captured by different representations when combined together can improve performance over individual models. However, we note that the Longformer-based Direct Prediction model has performed similarly well. It outperforms the rest of the Transformer-based pre-trained models that utilize only the first 512 tokens or legal summaries highlighting the beneficial effect of greater context to the task.

Looking at the 512-token limitation models (Direct Prediction and Summary-based Direct Prediction) (Table 9 there appears to be no clear model that performs better than the rest highlighting the sensitivity of the task. Of the Direct Prediction models, XLNet performs the best while in terms of summaries, RoBERTa using the KMM-based summaries shows the highest scores. In general KMM and LetSum-based summary models performed better as they use the rhetorical role information of the sentences to create their summaries. Models pre-trained on legal data did not show significant improvements in performance over their counterparts as the fact information used is mostly simple English and does not contain much law-specific language (Table 10).

While the Advocate-Area Joint Prediction model shows good performance, it is outperformed by the Longformer-based Direct Prediction model. Similarly, the Area-contextualised Representation-based model showcases poor performance for both variants. This degradation in performance when using area information is discussed in further detail in Section 7 and 7.2.

We implement a variant of the **doctor2vec** recommendation model (Biswal et al. 2020) to understand the ability of other expert recommendation models in predicting advocates (please refer to section 5.5 for details on this implementation). Our findings suggest that such a dynamic-databank strategy does not perform well. Pre-

Table 8 Implementation details for models used

Model	No. of parameters	No. of GPU hours	No. of epochs	Learning Rate
Longformer	149M	120	100	3e-5
BERT	110M	24	100	3e-5
LegalBERT	110M	24	100	3e-5
InlegalBERT	110M	24	100	3e-5
RoBERTa	123M	24	100	3e-5
XLNet	110M	24	100	3e-5
HAN	4.8M	8	200	1e-3

Table 9 Macro-performance of models for each category (in descending order of macro F1 scores)

Model	Precision	Recall	F1
Direct Prediction			
Longformer	0.3464	0.1558	0.1980
BERT (first 512 tokens)	0.3880	0.1419	0.1868
XLNet (first 512 tokens)	0.4582	0.1332	0.1856
InLegalBERT (first 512 tokens)	0.3465	0.1458	0.1839
RoBERTa (first 512 tokens)	0.3398	0.1392	0.1780
LegalBERT (first 512 tokens)	0.3608	0.1366	0.1795
HAN	0.2508	0.1084	0.1432
Summary-based Direct Prediction			
RoBERTa (KMM)	0.3522	0.1538	0.1912
BERT (LetSum)	0.4362	0.1377	0.1898
RoBERTa (LetSum)	0.3572	0.1467	0.1877
BERT (KMM)	0.3130	0.1426	0.1836
InLegalBERT (LetSum)	0.3828	0.1404	0.1834
LegalBERT (LetSum)	0.3682	0.1411	0.1832
InLegalBERT (KMM)	0.3488	0.1425	0.1831
RoBERTa (SummaRuNNer CNNRNN)	0.3196	0.1424	0.1797
XLNet (LetSum)	0.4292	0.1294	0.1790
BERT (SummaRuNNer CNNRNN)	0.3870	0.1352	0.1787
InLegalBERT (SummaRuNNer RNNRNN)	0.3524	0.1399	0.1786
BERT (CaseSummarizer)	0.3778	0.1349	0.1785
BERT (SummaRuNNer RNNRNN)	0.3885	0.1328	0.1785
LegalBERT(SummaRuNNer CNNRNN)	0.3595	0.1349	0.1779
LegalBERT(SummaRuNNer RNNRNN)	0.3346	0.1400	0.1776
LegalBERT (KMM)	0.3465	0.1373	0.1773
InLegalBERT (CaseSummarizer)	0.3642	0.1356	0.1771
InLegalBERT (SummaRuNNer CNNRNN)	0.3378	0.1309	0.1716
XLNet (KMM)	0.4401	0.1205	0.1716
LegalBERT (CaseSummarizer)	0.3629	0.1292	0.1709
XLNet (CaseSummarizer)	0.4150	0.1198	0.1669
XLNet (SummaRuNNer CNNRNN)	0.4118	0.1211	0.1666
XLNet (SummaRuNNer RNNRNN)	0.4133	0.1203	0.1664
Advocate-contextualised Representation-based			
Concatenation	0.4428	0.1780	0.2085
Area-contextualised Representation-based			
Area Representation (SBERT)	0.3320	0.1193	0.1598
Area Representation	0.2662	0.1063	0.1391
Joint Prediction			
Advocate-Area Joint Prediction	0.4070	0.1445	0.1904
Databank-based Prediction			
Databank (Biswal et al. 2020)	0.2851	0.1104	0.1414

trained language models perform far better than sequence-based models trained from scratch as is evidenced by the poor performance of the HAN model.

In Section 7, we delve into possible reasons why all models find the task challenging.

Table 10 Macro-performance of best performing models for each category (in descending order of macro F1 scores)

Model	Precision	Recall	F1
Direct Prediction			
Longformer	0.2931	0.1767	0.2014
XLNet (first 512 tokens)	0.3905	0.1446	0.1889
Summary-based Direct Prediction			
RoBERTa (KMM)	0.2682	0.1722	0.1916
Advocate-contextualised Representation-based			
Concatenation	0.4825	0.1557	0.2075
Area-contextualised Representation-based			
Area Representation (SBERT)	0.3320	0.1193	0.1598
Area Representation	0.2662	0.1063	0.1391
Joint Prediction			
Advocate-Area Joint Prediction	0.4070	0.1445	0.1904
Databank-based Prediction			
Databank (Biswal et al. 2020)	0.2851	0.1104	0.1414

7 Discussion

Given how challenging all the models find the task of advocate recommendation, we investigate the nature of advocates predicted. In particular, we consider some of the best performing members from each category as described in Section 5.

Analyzing the effect of training data on the ability of models to predict advocates better, we find that **models are not better at predicting advocates with more training data**. We find there to be a poor correlation between the label F1 score of an advocate and the number of training cases they possess (Column 'Corr. train' in Table 11). In addition, we find advocates with a class F1 score ≥ 0.75 to have less training data across all models (Column 'Avg. train' in Table 11). A correlation study between the co-occurrence levels of advocates and their class F1 scores reveals that advocates with more than 150 training cases have a consequently higher co-occurrence value which causes greater confusion for models and results in a lower prediction ability

Table 11 Per-advocate Prediction Analysis

Method	Corr. train	Avg. train	Corr. co-occur
Databank	0.2097	38.67	-0.5287
Longformer	0.1924	58.60	-0.6694
Concatenation	0.1871	42.34	-0.6914
Joint Prediction	0.1714	39.00	-0.6359
RoBERTa (KMM)	0.1599	38.67	-0.7553
XLNet (first 512 tokens)	0.1320	42.34	-0.7444
Area Prediction (SBERT)	0.1127	20.25	-0.5500

Column **Corr. train** is the Pearson correlation between the F1 score of each advocate and the number of training cases, Column **Avg. train** is the average number of training cases for advocates with a class F1 ≥ 0.75. Column **Corr. co-occur** is the Pearson correlation between the degree of the vertex of an advocate in the weighted co-occurrence graph and the F1 score of that advocate

(Column 'Corr. co-occur' in Table 11). Our analysis highlights that the strange scenario where advocates despite having more training data do not perform the best is due to the higher levels of co-occurrence they have. Therefore, **all models are good at predicting advocates with relatively low co-occurrence with other advocates**. For a detailed discussion, please refer to section 7.1.

The area-based models (Sections 5.3.2 and 5.4) suffer due to the ability of the area prediction models. Our area prediction model achieves an F1 score of 0.4328. We observe that the SOTA performance in the closely related area of Legal Statute Prediction is 0.566 (Paul et al. 2020) (uses several extra attributes that we do not possess and can therefore, not incorporate). The relatively poor performance of area and legal statute prediction is indicative of why the area-based models have degraded performance. Therefore, the area-based prediction strategies are limited by the ability of area and statute prediction models. As the SOTA performance in these areas improves, we hope that the area-based strategies will improve as well.

7.1 Prediction analysis

As there is a significant imbalance in the amount of training data for each advocate (Table 6, Section 7), we investigate whether models end up performing better for advocates with more data.

Column 'Corr. train' in Table 11 highlights the Pearson correlation between the number of training cases of an advocate and its class F1 score on the test set. All models show poor correlation with a maximum of 0.2097 for the doctor2vec-based Databank. This highlights that surprisingly, **models are not better at predicting advocates with more training data**. In fact, we find that the average number of training cases for advocates with an F1 score ≥ 0.75 to be quite low (Column 'Avg. train' in Table 11).

The same set of 7 advocates is predicted across all models with 2 of them featuring in all the considered models and 3 featuring in 5 of them. We observe that these advocates i.e. 'ShivChopra', 'ArtaTranaPanda', 'SMuralidhar', 'AjayKalra', 'AjayKumar', 'ManojSharma' and 'SatyenSethi' contain cases from the areas of 'CRIMINAL LAW', 'BANKING AND FINANCE LAW', 'CONSTITUTIONAL LAW' and 'INTERNATIONAL LAW'. The similarity in the best predicted advocates suggests that all models are able to accurately predict the cases for the same set of advocates.

Given this similarity across all models, we investigate further whether various models have differing abilities to predict different advocates. We compute the Spearman rank correlation between the per-advocate F1 scores for every pair of the models considered. We find that the similarity in advocates for high F1 scores is consistent throughout the range of F1 scores. The average Spearman rank correlation between any two pairs of models is 0.7998. This highlights that for the greater part, all models find the same advocates to be challenging or easy to predict correctly.

In order to understand why this deviation in expected behaviour occurs, we look at whether co-occurrence of advocates plays a significant role in their predictability. Our understanding of the performance is that models struggle to predict advocates that have a large amount of co-occurrence with other advocates. As a result, advocates with more training cases (therefore having more co-occurrence) do not achieve

the best F1 score. To verify this, we construct a weighted advocate co-occurrence graph with the nodes as the advocates and the edges between them being the number of training cases they feature together. We investigate the effect of label co-occurrence on the F1 scores of each advocate across all models. Column 'Corr. co-occur' in Table 11 highlights the correlation between the degree of advocates in the constructed graph with greater than 150 cases and their F1 for the considered models. We observe a relatively large amount of negative correlation between the two quantities across all models. This confirms that when advocates have a large amount of training data, they usually have a large amount of advocate co-occurrence which results in degradation of performance for those advocates. This is further reinforced by the fact that we did not observe similar correlation when considering advocates having less than 150 cases (average Pearson correlation: -0.1265). We find that most of the advocates with the best class F1 scores across all the models have fairly low degrees of co-occurrence.

Our observations suggest that **models are good at predicting advocates with relatively low co-occurrence with other advocates**. As advocates feature in more cases, their co-occurrence values increase. As a consequence advocates with more training cases do not necessarily have the best F1 scores.

7.2 Area-based analysis

Using the area annotation as described in Section 4.2, we investigate if models are able to recommend advocates belonging to certain areas better than the rest.

To understand this, we recompute the F1 scores of each advocate for every area by considering only the test cases of that area. We then compute the macro F1 score for that area by averaging over the F1 scores of the advocates with cases in that area. We find the areas with the best F1 scores to be similar across all the methods and to have very few training cases (≤ 20) These areas are: 'MARITIME LAW', 'MINING, PETROLEUM AND NATURAL RESOURCES LAW', 'MEDIA LAW', 'CONSUMER LAW', 'TELECOMMUNICATION LAW' and 'IMMIGRATION LAW'. We observe that the areas with the highest F1 scores have very few training cases (≤ 20). Also, while the advocates report relatively high F1 scores for each area, their overall F1 score is not seen to be very high (≤ 0.48). This highlights that areas play a contributory role to the difficulty of predicting advocates and models find it easier to predict advocates for cases from particular areas. However, performing a co-occurrence analysis similar to Section 7 did not provide any significant trends. Therefore, while areas do play a significant role in controlling the ease or difficulty of predicting advocates, this effect is not at easily understood and requires further work.

The poor performance of the area-based models can be attributed to the quality of area prediction. Our Longformer-based area prediction model achieves an F1 score of 0.432 and we find that the SOTA performance in the closely related area of Legal Statute Prediction to be relatively low as well i.e. 0.566 (Paul et al. 2020). Relatively poor performance on area prediction results in poor area embeddings which causes advocate prediction to suffer. The degradation in the performance of the Advocate-Area Joint Prediction model (Section 5.4) can similarly be attributed to the poor performance of the area prediction model. Thus, while advocates have

clear areas of expertise, poor prediction quality of the areas of new facts leads to a drop in performance when incorporating area information while recommending advocates. We believe, that as the SOTA performance of the related area of Legal Statute Prediction, improves, using area information in advocate prediction will lead to better performance than direct prediction.

While advocates have areas of expertise, considering the prediction of any advocate belonging to a particular area for the area of a case cannot be considered as realistic for a few reasons. First and foremost, cases related to large areas like 'CRIMINAL LAW' dominate the dataset. For such areas, we find that almost all of the 306 advocates would be considered relevant and therefore correct. As a result, this would make the task much too permissive and unrealistic. More importantly, given the relatively poor performance of area prediction, relying solely on the predicted areas of a case to recommend advocates might lead to very incorrect recommendations. Lastly, even though advocates can have similar areas of expertise, there may be subtle variations in them that cannot be captured by the area granularity. In such instances, considering all advocates within the same area as equivalent would be incorrect.

8 Conclusion and future work

In this paper, we propose a socially pertinent and practical task – advocate recommendation, in the Indian legal scenario.

To promote research on this task, we also contribute a new dataset named "ARDI" (see Section 4) with information of 300+ advocates. In addition, the dataset is rich in information like per-document statute information, expert-annotated area information, per-sentence level rhetorical role information, and verdict information.

We perform comprehensive benchmarking on the ARDI dataset with many machine learning models (see Section 5) presented as elaborate results (Section 6.3), followed by detailed analysis (see Section 7).

Through extensive experimentation with a wide variety of SOTA pre-trained models, summarization, representation-based, and joint learning strategies, we highlight the challenging nature of the task. Our observations indicate that using greater context length and ensembling information over a large number of models proves beneficial to the task. However, area incorporation into our models leads to degradation of performance on account of poor area predictability. Contrary to expectations, advocates with more training data were not easier to predict due to greater co-occurrence. In the future, we look to explore avenues of incorporating the winning abilities of different advocates. We will also look to investigate the effects of minimizing data imbalances inherent in the dataset as well as focus on improving performance by incorporating area information.

9 Limitations

While our work considers a realistic problem and approaches it from a practical perspective, there is room for improvement. The present work considers advocates extracted from legal documents. It is unable to account for advocates that are unavailable or have retired from the legal practice. As this information is unable, it is also unable to account for any monetary transactions involved in taking on clients. The facts we work with are not gold-standard, instead having been extracted by a model which itself introduces inaccuracies. Lastly, while we consider models that work with longer sequences, we are aware that several documents still far exceed these token lengths. We look to remedy these limitations in future works and welcome suggestions and contributions from the community for the same.

In the Indian system, although there is a pre-trial phase, the advocates do not always have a substantial role to play there to keep the trial away from the court. The concepts of pre-trial mediation (court-mandated or voluntary), negotiation and settlement, while not entirely unknown in India, are yet to gather any significant momentum except for in high-profile corporate litigation involving significantly high stakes. The major reason why all civil disputes may not reach a trial stage in India is the lack of resources and awareness among the masses about available rights and remedies. Therefore, while the ability to arrive at a pre-trial settlement may be considered a valuable trait in a lawyer, that may not prove to be a determinant when it comes to the more common practice of hiring the services of an advocate to represent oneself before a court of law. Further, there are transactional lawyers and litigators (advocates), who are often clearly if informally demarcated in the Indian legal industry, and the former, though heavily outnumbered by the latter, are those who tend to engage in such pre-trial activities rather than the litigators, who focus on courtroom results. In addition, in criminal trials, plea bargaining is also conducted through the court with the judge playing a more involved role compared to the US justice system. Moreover, the only data publicly available and accessible in India are the final court orders, and not the plaints and written statements filed by the parties from the filing of the case leading to the final judgment. Little or no documentary record exists of the details of any negotiation or settlement reached during the pre-trial phase unless the parties choose to get the approval of the court for any settlement reached between them. This has handicapped us to only formulate the problem centered around the proficiency of the advocates in the final judgement since there is no recorded version of the lawyer's performance in the pre-trial phase. For the aforesaid reasons, we would like to assert that pre-trial negotiation ability in a lawyer, while an impactful factor in the American system, may not be considered relatively crucial in the Indian context, especially when compared to the ability to obtain a court order in favour of the client.

Declarations

Ethics Statement We do not make use of any publicly available information about the advocates or the areas. As the laws of India prohibit advocates advertising themselves, all information about advocates is extracted and inferred from the documents itself. Additionally, information about areas is only obtained from the legal experts through our annotation process. Therefore, we have adhered to the ethics of not using any self-declared advertised information. All cases in our dataset are obtained from Indian Kanoon using a paid API. The website was paid the requisite fees in keeping with ethical standards to facilitate downloading the cases' corresponding HTML documents. The legal experts were adequately compensated for their services on mutual agreement of the amount.

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